Chapter 9

Structure of finite dimensional algebras

In this chapter we return to studying the structure of finite dimensional algebras. Throughout the chapter, we work over an algebraically closed field k (of any characteristic).

9.1. Lifting of idempotents

Let A be a ring, and let $I \subset A$ be a nilpotent ideal.

Proposition 9.1.1. *Let $e_0 \in A/I$ be an idempotent, i.e., $e_0^2 = e_0$. There exists an idempotent $e \in A$ which is a lift of e_0 (i.e., it projects to e_0 under the reduction modulo I). This idempotent is unique up to conjugation by an element of $1 + I$.*

Proof. Let us first establish the statement in the case when $I^2 = 0$. Note that in this case I is a left and right module over A/I. Let e_* be any lift of e_0 to A. Then $e_*^2 - e_* = a \in I$, and $e_0 a = a e_0$. We look for e in the form $e = e_* + b$, $b \in I$. The equation for b is $e_0 b + b e_0 - b = -a$.

Set $b = (1 - 2e_0)a$. Then

$$e_0 b + b e_0 - b = -2e_0 a - (1 - 2e_0)a = -a,$$